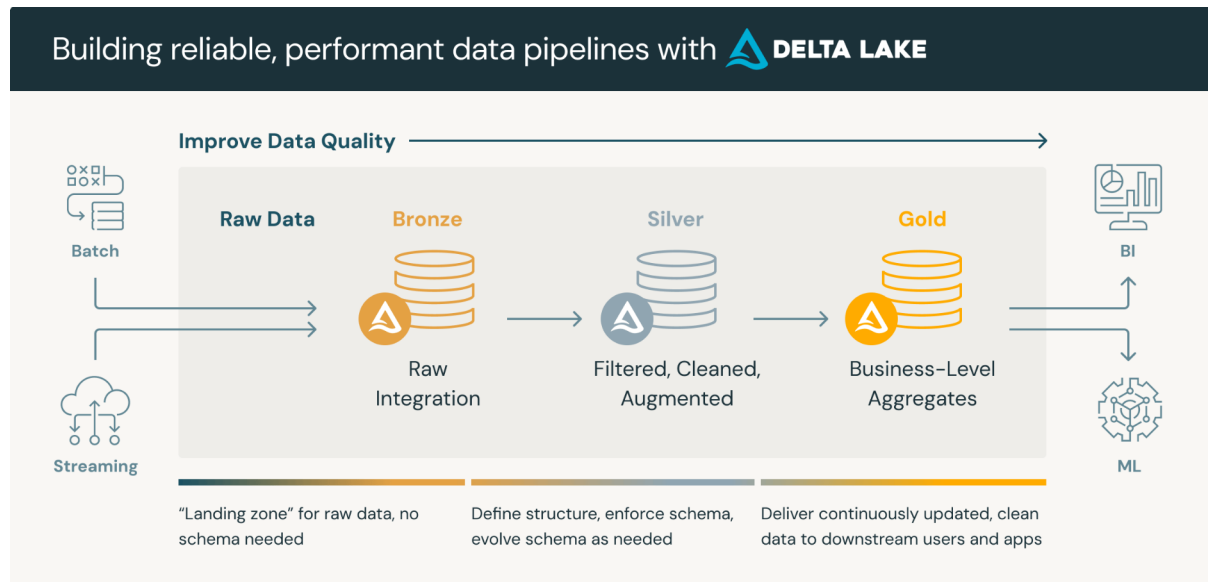


Architecture & Design

1. Architecture Overview

This project implements a **modern Lakehouse architecture** on Azure to support scalable, reliable, and analytics-ready data processing for a retail enterprise. The architecture follows the **Medallion pattern** (Bronze → Silver → Gold) to progressively refine raw data into high-quality analytical datasets.



2. High-Level Architecture Flow

The end-to-end data flow is as follows:

1. Raw sales and master data files are ingested into **Azure Data Lake Storage (ADLS)**.
2. Data is loaded into the **Bronze layer** in raw format using Apache Spark.
3. The **Silver layer** performs data cleansing, validation, deduplication, and calibration.
4. The **Gold layer** contains business-ready aggregated datasets.
5. **Power BI** connects directly to Gold Delta tables for reporting and visualization.

Source CSV Files



Azure Data Lake Storage



Bronze Layer (Raw Delta Tables)



Silver Layer (Cleaned & Validated Delta Tables)



Gold Layer (Aggregated Analytics Tables)



Power BI Dashboards

3. Technology Stack & Roles

Azure Databricks

Azure Databricks is used as the primary data engineering platform to:

- Execute distributed data processing jobs
- Orchestrate the Bronze → Silver → Gold pipeline
- Manage notebooks and workflows

Apache Spark

Apache Spark is the core processing engine responsible for:

- Large-scale data ingestion
- Data transformations and aggregations
- Incremental and batch processing using PySpark and Spark SQL

Delta Lake

Delta Lake provides reliability and performance by enabling:

- ACID transactions on data lake storage
- Incremental UPSERTs using MERGE
- Schema evolution
- Time travel and rollback capabilities

Power BI

Power BI is used for analytics and reporting:

- Connects directly to Gold Delta tables
- Supports interactive dashboards and scheduled refresh
- Enables business users to analyze trends and KPIs

4. Medallion Architecture Design

The project follows the Medallion Architecture pattern:

Layer	Description	Data State
Bronze	Raw data ingestion	Uncleaned, unvalidated
Silver	Cleansed and validated data	Trusted, structured
Gold	Aggregated business metrics	Analytics-ready

Bronze Layer

- Stores raw data in Delta format
- Preserves original schema
- Adds ingestion metadata
- Supports file-based incremental ingestion

Silver Layer

- Cleans and validates data
- Removes duplicates
- Applies business rules and calibration
- Separates invalid records into a quarantine table

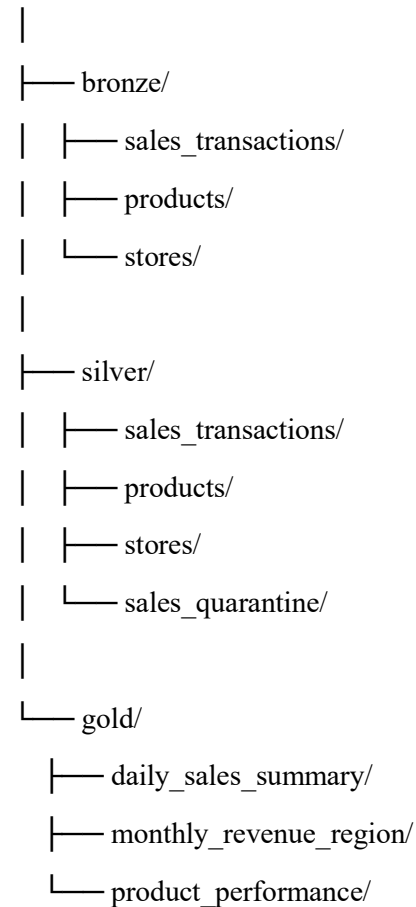
Gold Layer

- Stores aggregated, analytics-ready data
- Optimized for reporting
- Registered as managed tables in Unity Catalog

5. Data Lake Storage Structure

The Azure Data Lake is organized into clear layer-based directories:

/Volumes/main/default/retail_datalake/



6. Partitioning Strategy

Partitioning is applied to optimize performance and scalability.

Bronze Layer

- Partitioned by ingestion_date
- Enables efficient incremental ingestion and storage management

Silver Layer

- Partitioned by transaction_date
- Optimizes filtering and downstream transformations

Gold Layer

- Partitioned by year and month
- Optimized for time-based analytics and Power BI queries

7. Data Flow Summary

- **Source** → **Bronze**: Raw ingestion with metadata
- **Bronze** → **Silver**: Cleansing, validation, deduplication, quarantine handling
- **Silver** → **Gold**: Business aggregations and KPI creation
- **Gold** → **Power BI**: Analytics and visualization

8. Architecture Benefits

- Scalable and cloud-native
- Fault-tolerant and ACID compliant
- Supports incremental processing
- Optimized for enterprise BI reporting
- Clear data lineage and governance